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Project title: Student skills marketplace

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List of tables

Table 2.1 Users of the current system

Table 3.1 Actors of the proposed system

Table 3.2 Register as service provider usecase description

Table 3.3 Service creation usecase

Table 3.4 Order placing usecase

Table 3.5 Deliver completed work use case

Table 3.6 leave review and rating usecas

List of figures

Figure 1.1 project schedule

Figure 2.1 use case diagram of the existing system

Figure 3.1 Use case diagram for the proposed system

Figure 3.2 Class diagram

Figure 3.3 Object diagram

Figure 3.4 Sequence diagram for login usecase

Figure 3.5 sequence diagram for service listing creation

Figure 3.6 sequence diagram for order placement

Figure 3.7 activity diagram

Figure 3.8 state diagram

Figure 4.1 system decomposition

Figure 4.2 component diagram

Figure 4.3 deployment diagram

Figure 4.4 Entity Relation Diagram

figure 4.5 users table

figure 4.6 services table

figure 4.7 orders table

figure 4.8 messages table

figure 4.9 payments table

figure 4.10 reviews table

Table of contents

List of tables	2
List of figures	3
CHAPTER ONE	6
1.1 Introduction	6
1.2 Background of the project	6
1.3 Statement of the project	7
1.4 Purpose of the project	8
1.5 Objective of the project	8
1.5.1 general objective	8
1.5.2 specific objective	8
1.6 Feasibility study	9
1.6.1 Technical feasibility	9
1.6.2 Operational feasibility	9
1.6.3 Economic fallibility	9
1.7 Scope and limitation	10
1.7.1 Scope	10
1.7.2 limitation	10
1.8 Significance of the project	11
1.9 Methodology	12
1.9.1 Fact finding methodology	12
1.9.2 Software Development Methodology	12
1.10 Development tools	13
1.11 Testing plan	13
1.11.1 unit test	13
1.11.2 Integration test	13
1.12 Overview of the project phase	14
1.12.1 schedule	14
1.12.3 Team composition	14
CHAPTER TWO: Overview of the current system	15
2.1 Major functionality of the current system	15
2.2. User of the current system	15

2.3. Problem of the current system	17
2.4. Business rule of the current system	17
CHAPTER THREE: Proposed system	18
3.1 Overview	18
3.2 Functional requirement	19
3.3 Non functional requirement	20
3.4 System model	21
3.4.1 use case model	21
3.4.1.1 actor identification	21
3.4.1.2 use case diagram and description	22
3.4.2 Static model	26
3.4.2.1 class diagram	26
3.4.2.1 object diagram	27
3.4.3 Dynamic model	27
3.4.4.1 sequence diagram	27
3.4.4.2 activity diagram	30
3.4.4.3 state diagram	31
CHAPTER FOUR: System design	32
4.1 Overview	32
4.2 Purpose of the system design	32
4.3 Design goal	32
4.4 proposed system architecture	33
4.5 system process	34
4.6 subsystem decomposition	36
4.7 persistent data management	36
4.8 component diagram	37
4.9 deployment diagram	37
4.10 database design	38
4.10.1 conceptual database design	38
4.10.2 physical database design	39
4.11 access control	41
4.12 user interface design	42
4.13 reference	42

CHAPTER ONE

1.1 Introduction

The Student Skill Marketplace is a web-based platform designed to connect university students in Ethiopia with potential clients seeking their services. This project aims to create a structured digital environment where students can offer their skills, build portfolios, and generate income while clients can easily discover, evaluate, and engage with student talent. In today's digital economy, specialized platforms have transformed how services are exchanged, yet Ethiopian students lack dedicated infrastructure to market their skills effectively. This project addresses this gap by developing a comprehensive marketplace solution tailored to the unique needs of student service providers and their clients.

The platform will leverage modern web technologies including PHP, MySQL, HTML with Tailwind CSS, and JavaScript to create an intuitive, secure, and efficient marketplace. By incorporating Stripe payment processing, the system will ensure transparent and protected financial transactions for all users. This introduction chapter outlines the project's background, objectives, scope, and implementation approach, establishing the foundation for the development of a solution that empowers Ethiopian students to monetize their skills while providing clients with access to affordable, quality services.

1.2. Background of the Project

In recent years, the gig economy has revolutionized how work is performed and services are delivered globally. Platforms like Fiverr, Upwork, and Freelancer have created opportunities for professionals to offer their skills to a global market. However, there remains a significant gap in the Ethiopian context, particularly for university students who possess valuable skills but lack dedicated platforms to monetize them. Ethiopian students often have remarkable technical, creative, and academic abilities acquired through their education, yet face challenges in connecting with potential clients who might benefit from these skills.

The current economic landscape in Ethiopia presents unique challenges for students. The rising cost of education, coupled with limited part-time employment opportunities, creates financial strain for many students pursuing higher education. Concurrently, businesses, startups, and individuals in Ethiopia often require specialized services that students could

effectively provide at competitive rates. This mismatch between student skills and market needs represents an untapped economic opportunity.

The Student Skill Marketplace project aims to address this gap by creating a web-based platform specifically designed for Ethiopian students to offer their services to potential clients. This platform will facilitate connections between skilled students and clients seeking their expertise, while providing secure payment processing through Stripe checkout and Stripe connect integration.

1.3. Statement of the Problem

The current situation in Ethiopia presents several challenges for both students and potential clients:

- Students with valuable skills lack a dedicated platform to market their services to potential clients
- Traditional job-seeking methods often fail to accommodate the flexible schedules students require
- Businesses and individuals seeking specialized services have limited access to the student talent pool
- The absence of a trusted intermediary creates concerns about payment security and service quality
- Manual service discovery and client acquisition processes are time-consuming and inefficient
- Students struggle to establish credibility and build portfolios without a structured platform
- Payment processing remains challenging without specialized systems for service-based transactions
- The fragmented marketplace makes it difficult for students to determine competitive pricing
- Potential clients cannot easily compare service providers or verify their qualifications
- Students lack exposure to real-world professional experiences before graduation
- These challenges collectively hinder the potential economic exchange between skilled students and clients who need their services, resulting in missed opportunities for both parties.

1.4. Purpose of the Project

The purpose of the Student Skill Marketplace project is to develop a comprehensive web-based platform that connects skilled university students in Ethiopia with clients seeking their services. The platform will create a structured ecosystem where students can showcase their abilities, build portfolios, and generate income while clients can discover, evaluate, and engage student talent efficiently. By implementing Stripe payment integration, the platform will ensure secure, transparent financial transactions that build trust in the marketplace.

The platform aims to foster an environment where students can gain practical experience, develop entrepreneurial skills, and supplement their income while studying. For clients, the marketplace will provide access to a diverse pool of affordable talent with specialized skills that might otherwise be difficult to source. The system will manage the entire service lifecycle from discovery to delivery and payment, creating a seamless experience for all users.

1.5. Objective

1.5.1. General Objective

The general objective of this project is to design and implement a secure, efficient web-based Student Skill Marketplace platform that enable students to monetize their skills.

1.5.2. Specific Objective

The following are the specific objectives of the project that are identified in line with achieving the general objective:

- Design and implement a user-friendly platform with separate interfaces for students and clients
- Create a service listing module that allows students to showcase their skills with appropriate categorization
- Implement a robust order and booking management system to facilitate service delivery
- Integrate Stripe Connect to enable secure payment processing and funds distribution to students
- Develop a messaging system that enables clear communication between students and clients

- Implement a rating and review system to establish trust and credibility in the marketplace
- Create notification systems to keep users informed about relevant activities

1.6. Feasibility Study

1.6.1. Technical Feasibility

The Student Skill Marketplace project is technically feasible based on the following considerations:

- The required technologies (PHP, MySQL, HTML, Tailwind CSS, JavaScript) are well-established with extensive documentation and community support
- Stripe API provides comprehensive documentation for payment processing integration
- The project architecture follows standard web application design patterns with proven scalability
- Hosting environments for PHP/MySQL applications are widely available and affordable
- Development tools and environments for the selected technology stack are mature and reliable
- The required security implementations for user data and payment processing are established practices
- The system architecture can accommodate future expansions and feature additions

1.6.2. Operational Feasibility

From an operational perspective, the project is feasible based on these factors:

- The functionality aligns with existing user behaviors but addresses key pain points in the current informal system
- The platform's intuitive design will minimize the learning curve for both students and clients
- Similar marketplace platforms have proven operational success in other contexts
- The system will include comprehensive user guidance and support documentation
- Students and clients are increasingly adopting digital solutions for service exchange
- The marketplace model is well-understood by the target demographic
- Administrative requirements for platform maintenance are reasonable and manageable

1.6.3. Economic Feasibility

The economic feasibility of the project is supported by the following analysis:

- Development costs are justified by the potential revenue through platform commission fees
- The marketplace model creates sustainable revenue through transaction percentages
- Initial investment requirements are reasonable compared to potential returns
- The platform addresses a significant market gap with demonstrated demand
- Operating costs post-launch are proportional to transaction volume, ensuring scalability
- The potential user base (university students in Ethiopia) represents a significant market
- The project leverages cost-effective open-source technologies where possible
- The platform can begin generating revenue immediately upon launch through transaction fees
- Return on investment is projected within an acceptable timeframe based on user adoption estimates

1.7. Scope and Limitation

1.7.1. Scope

The system will support registration, authentication, and profile management for students (service providers), clients, and administrators with appropriate role-based access control. Students will be able to create, edit, and manage service listings with descriptions, pricing, delivery timelines, and sample work. The platform will provide search, filter, and discovery features to help clients find appropriate services based on categories, skills, ratings, and price ranges. An integrated messaging system will enable pre-order and post-order communication between students and clients within the platform. Stripe integration will handle secure payment collection, holding funds in escrow until service completion, and disbursement to student providers after deducting platform commission. After service completion, clients can rate and review student providers to build platform credibility. Email and in-app notifications will keep users informed about orders, messages, and platform updates.

1.7.2 Limitation

The platform will initially support only English as the primary language of communication.

Payment functionality will be limited to Stripe's supported payment methods and currencies. The platform will not include video conferencing tools; users must rely on external solutions for video meetings. Initial deployment will be limited to web browsers with progressive mobile responsiveness rather than native mobile applications. Verification of students status isn't available. The platform will not implement Ethiopian tax collection or reporting mechanisms. The system will not incorporate verification with government business licensing requirements for service providers.

1.8. Significant of the Project

The Student Skill Marketplace offers significant benefits to various stakeholders:

For Students:

- Provides a structured platform to monetize existing skills and generate income while studying
- Creates opportunities to build professional portfolios with verified client work
- Develops entrepreneurial, communication, and client management skills through practical experience
- Offers flexible work opportunities that accommodate academic schedules
- Establishes a professional reputation through ratings and reviews
- Reduces financial pressure through supplementary income opportunities

For Clients:

- Provides access to affordable skilled services from university students
- Simplifies the process of finding and hiring specialized talent
- Ensures payment security through escrow functionality
- Enables quality assessment through ratings and portfolio reviews
- Supports local talent development while addressing service needs
- Offers transparent pricing and delivery expectations

For Educational Institutions:

- Enhances students' practical application of theoretical knowledge
- Creates pathways for students to develop professional skills beyond the classroom
- Potentially reduces student dropout rates due to financial pressures

- Strengthens connections between academic learning and marketplace needs
- Provides data on in-demand skills that can inform curriculum development

For the Ethiopian Economy:

- Stimulates economic activity by facilitating service transactions
- Encourages entrepreneurship and self-employment among youth
- Reduces barriers to market entry for service providers
- Promotes skill development aligned with market demands
- Creates an ecosystem for knowledge and skill exchange

1.9. Methodology

1.9.1. Fact Finding Methodology

The project team will employ the following methods to gather requirements and understand the needs of all stakeholders:

- Structured interviews with university students to identify their marketable skills and challenges in monetizing them
- Surveys among potential clients to understand service needs and expectations
- Focus group discussions with both students and potential clients to validate initial findings
- Competitor analysis of existing freelance platforms to identify best practices and opportunities for differentiation
- Document analysis of relevant research papers and market reports on gig economy platforms
- Observation of current informal student service provision methods to understand existing workflows
- Expert consultations with faculty members and industry professionals regarding skill marketability

1.9.2. Software Development Methodology

The project will adopt an Agile development approach with Scrum framework to ensure flexibility, stakeholder involvement, and iterative delivery:

- Planning: Define project scope, requirements, and acceptance criteria

- Sprint Reviews: Demonstrate completed features to stakeholders for feedback
- Sprint Retrospectives: Identify process improvements for subsequent sprints
- Continuous Integration: Implement automated testing and deployment pipelines
- User Testing: Involve potential users in testing sprints to validate usability

1.10. Development Tools

- Frontend: HTML5 with Tailwind CSS, vanilla JavaScript (with optional frameworks as needed)
- Backend: Vanilla PHP for server-side processing
- Database: MySQL for relational data storage
- Package Management: Composer for PHP dependency management
- Payment Processing: Stripe API for payment handling and Connect for marketplace payments
- Authentication: PHP session management with proper security measures
- Version Control: Git with GitHub for code management
- Server: Apache web server with proper configuration for security

1.11. Testing Plan

1.11.1. Unit Test

Unit tests will be implemented for all major components to verify that individual parts of the application function correctly in isolation. These tests will cover:

- User authentication and authorization functionality
- Service listing creation and management
- Order processing workflows
- Payment processing integration
- Messaging system functionality
- Rating and review mechanisms
- Notification generation and delivery

Unit tests will be automated using PHPUnit and run as part of the continuous integration pipeline to ensure code quality and prevent regression issues.

1.11.2. Integration Test

Integration tests will verify that different components of the system work together correctly, focusing on:

- Authentication system integration with user profiles
- `Service listing integration with search and discovery features
- Order management integration with payment processing
- Notification system integration with user actions
- Admin dashboard integration with platform data
- Stripe Connect integration with the marketplace payment flow

These tests will validate that data flows correctly between system components and that interdependent features function as expected.

1.12. Overview of Project Phase

1.12.1. Schedule

The project will be executed over an estimated period with the following timeline:

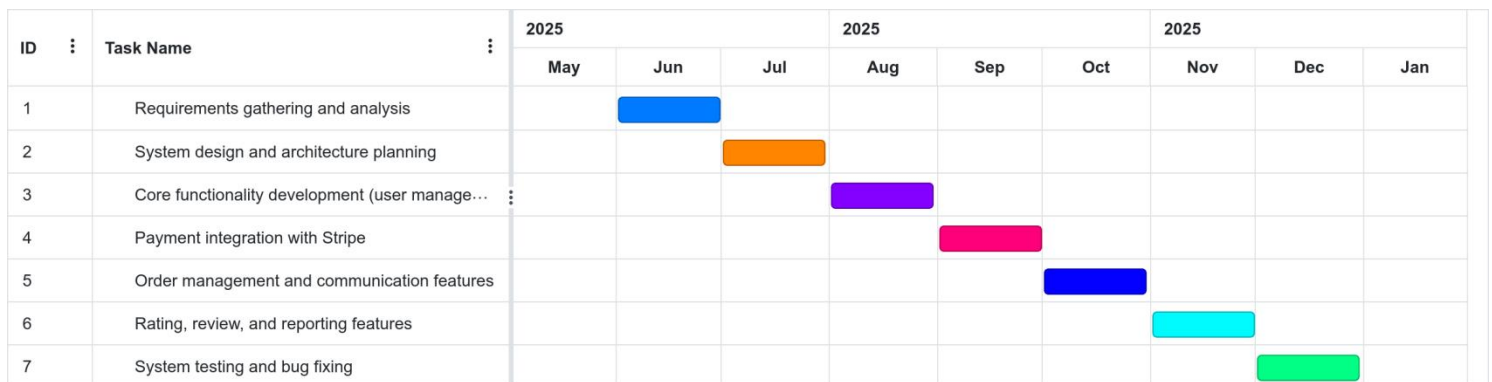


figure 1.1 project schedule gantt chart

1.12.2. Team Composition

The project team will consist of the following roles:

- Project Lead: Responsible for overall project coordination and stakeholder communication
- PHP Developer: Implementation of server-side logic and database interactions
- Frontend Developer: Implementation of user interfaces using HTML, Tailwind CSS, and JavaScript
- Database Administrator: Design and optimization of MySQL database architecture
- UI/UX Designer: Creation of user-friendly interfaces and user experience flows

- System Administrator: Setup and maintenance of development and deployment infrastructure

CHAPTER TWO: OVERVIEW OF CURRENT SYSTEM

2.1. Major functionality of the current system

The current system for connecting students with potential clients in Ethiopia operates as an informal and fragmented arrangement without a dedicated platform. Students typically use various general-purpose communication channels and personal networks to offer their services. The major functionalities of this current system include:

- Students advertise their services through social media platforms like Facebook, Telegram groups, and WhatsApp
- Students create basic portfolios on free platforms to showcase their previous work samples
- Students and clients connect through word-of-mouth referrals from peers, professors, and previous clients
- Service terms, pricing, and deliverables are negotiated through direct messaging or in-person meetings
- Service delivery occurs through email, file-sharing services, or in-person handovers
- Payments are processed through cash transactions, bank transfers, or mobile money services
- Students collect testimonials from satisfied clients informally via text messages or social media comments
- Students maintain personal client lists and communication records using basic tools like spreadsheets or notebooks

These functionalities enable a basic level of service exchange but lack standardization, security features, and scalability that a dedicated platform would provide.

2.2. User of the current system

The current informal system involves several categories of users, each with specific roles and interactions:

Users of the Current system

Users	Description
-------	-------------

Students	University students who offer various services based on their skills and academic knowledge. They manage the entire service process from promotion to delivery and payment collection.
Clients	Individuals, small businesses, and organizations seeking services who must search through fragmented channels to find appropriate student providers.
Intermediaries	Other platforms like social medias, friends, family members, and former clients who facilitate connections between students and potential new clients through personal recommendations.

Table 2.1: users of the current system

Each user category faces unique challenges in the current system. Students struggle with limited reach and credibility issues, while clients face difficulties in finding and evaluating potential service providers. Both parties lack security mechanisms for transactions and standardized processes for service delivery.

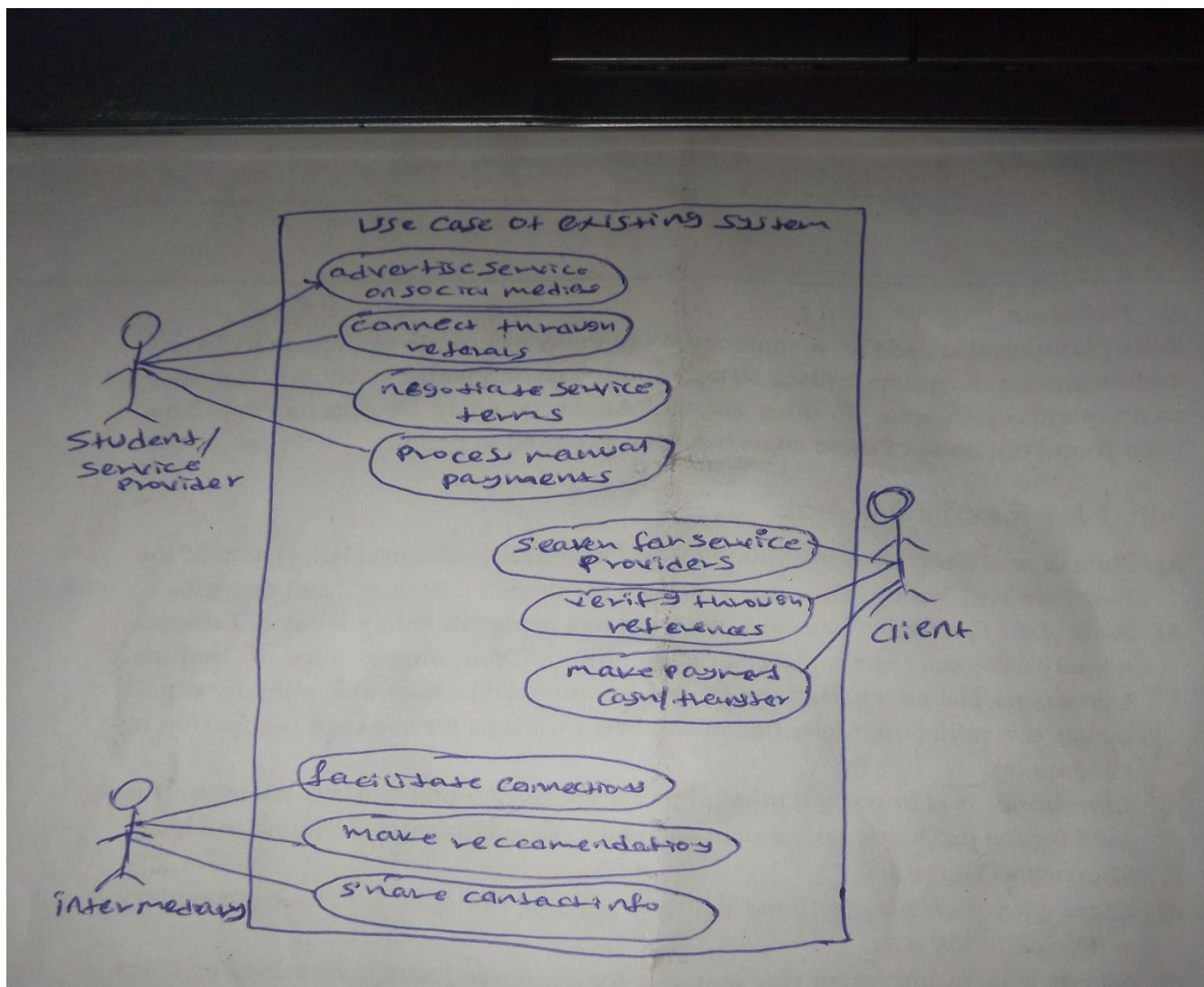


Figure 2.1 use case diagram of the existing system

2.3. Problem of the current system

The current informal approach to student service provision presents numerous problems that limit effectiveness for all participants:

- **Fragmented Discovery:** Students and clients struggle to find each other across disconnected platforms and networks, resulting in missed opportunities for both parties
- **Limited Market Reach:** Students can only effectively market their services to their immediate networks, significantly restricting their potential client base
- **Pricing Inconsistency:** Without market transparency, students lack reference points for appropriate pricing, often resulting in undervaluing their services
- **Payment Insecurity:** Direct cash or transfer payments offer no protection against non-payment for completed work or advance payment without delivery
- **Trust Deficit:** Clients have limited means to verify student qualifications or past performance beyond unverified testimonials
- **Communication Challenges:** Conversations scattered across various messaging platforms create disorganization and difficulty tracking service-related discussions
- **Absence of Standards:** Service descriptions, delivery expectations, and quality parameters vary widely without platform-enforced standards
- **No Dispute Resolution:** When conflicts arise over quality, payment, or deliverables, there is no formal system for resolution
- **Portfolio Limitations:** Students lack a structured way to build and showcase comprehensive portfolios of verified work
- **Scheduling Complexity:** Managing service delivery alongside academic responsibilities becomes challenging without dedicated project management tools
- **Data Security Concerns:** Sensitive client information and intellectual property are exchanged without proper confidentiality protections
- **No Performance Metrics:** Students receive limited structured feedback to improve their services over time

These problems collectively reduce efficiency, limit market potential, and create unnecessary risks for both students and clients in the service exchange process.

2.4. Business rule of the current system

Despite its informal nature, the current system operates according to several implicit business rules that govern interactions between students and clients:

Business rule one

Payment arrangements are established through verbal agreements or messaging platform discussions between students and clients. Payment timing may be structured as advance payment, completion-based payment, or milestone-based payment depending on mutual agreement, with no standardized payment protection mechanisms in place.

Business rule two

Late delivery consequences are handled through informal negotiation between student and client. No standardized penalties exist for missed deadlines.

Business rule three

In cases where a student receives advance payment or deposit and subsequently becomes unresponsive or fails to deliver services (ghosting), clients have no formal recourse mechanism for payment recovery.

Business rule four

Student credibility and reputation are established exclusively through personal referrals and recommendations from previous clients,. No formal rating or verification system exists to validate student qualifications or service quality.

CHAPTER THREE: PROPOSED SYSTEM

3.1. Overview

The proposed Student Skill Marketplace system is a web-based platform designed to connect Ethiopian university students with potential clients seeking their skills and services. The system will be developed using PHP for server-side processing, MySQL for database management, HTML with Tailwind CSS for frontend presentation, and JavaScript for client-side functionality. It will integrate Stripe's payment processing system to handle secure transactions between clients and student service providers.

The platform will operate as a C2C (Consumer-to-Consumer) marketplace where students can register, create service listings, communicate with clients, fulfill orders, and receive payments. Clients can browse available services, place orders, communicate requirements, review deliverables, and process payments. The system will provide robust search and

filtering capabilities, secure user authentication, messaging functionality, order management, payment processing, and a review system to build marketplace trust.

The architecture follows a modular design with clear separation between presentation, application logic, and data layers. This approach ensures maintainability, scalability, and security while providing a responsive user experience across devices. The system will be deployed on an Apache web server with appropriate security configurations to protect user data and financial transactions.

3.2. Functional Requirement

The proposed Student Skill Marketplace system must fulfill the following functional requirements:

1. User Accounts & Authentication

- 1.1 The system shall allow students to register as service providers with email verification
- 1.2 The system shall allow clients to register with email verification
- 1.3 The system shall provide secure login/logout functionality for all users
- 1.4 The system shall implement role-based access control (admin, student, client)
- 1.5 The system shall allow users to reset passwords securely
- 1.6 The system shall allow users to manage their profile information and profile pictures

2. Service Listings

- 2.1 The system shall allow students to create service listings with title, description, price, delivery time, and category
- 2.2 The system shall allow students to edit or delete their service listings
- 2.3 The system shall allow students to upload sample work as attachments to listings
- 2.4 The system shall allow students to specify delivery timeframes for each service
- 2.5 The system shall allow students to categorize services based on predefined categories
- 2.6 The system shall allow students to add tags to increase service discoverability

3. Marketplace Browsing

- 3.1 The system shall provide search functionality by keywords, category, and price range
- 3.2 The system shall allow filtering of services based on multiple criteria
- 3.3 The system shall implement pagination for service listing results
- 3.4 The system shall display student ratings and reviews on service listings
- 3.5 The system shall allow sorting of service listings by various criteria (price, rating, etc.)

4. Order Management

- 4.1 The system shall allow clients to place orders for services
- 4.2 The system shall allow clients to provide detailed requirements when placing orders
- 4.3 The system shall allow file uploads for requirements and deliverables
- 4.4 The system shall allow clients to request revisions to delivered work
- 4.5 The system shall facilitate order completion and mark orders as finished

5. Payment Processing

- 5.1 The system shall integrate Stripe Connect for secure payment processing
- 5.2 The system shall hold client payments in escrow until order completion
- 5.3 The system shall automatically transfer funds to students upon order completion
- 5.4 The system shall deduct platform commission fees from transactions
- 5.7 The system shall store transaction records securely

6. Messaging & Communication

- 6.1 The system shall provide an in-platform messaging system between students and clients
- 6.2 The system shall allow file attachments in messages
- 6.3 The system shall notify users of new messages via email and in-app notifications
- 6.4 The system shall maintain message history for all conversations

7. Ratings & Reviews

- 7.1 The system shall allow clients to rate students on completed orders
- 7.2 The system shall allow clients to leave written reviews for students
- 7.3 The system shall calculate and display average ratings for students
- 7.4 The system shall allow students to respond to client reviews
- 7.5 The system shall display recent reviews on student profiles

8. Admin Dashboard

- 8.1 The system shall provide administrators with a comprehensive dashboard
- 8.2 The system shall allow administrators to manage users (view, suspend, delete)
- 8.3 The system shall allow administrators to manage service listings
- 8.4 The system shall allow administrators to manage categories and tags
- 8.5 The system shall provide system-wide analytics and reports

3.3. Non-functional Requirement

The proposed system must fulfill the following non-functional requirements:

1. Performance

- 1.1 The system shall load pages faster under normal load conditions

1.2 The system shall maintain responsiveness during peak usage periods

2. Security

2.1 The system shall encrypt all sensitive data stored in the database

2.2 The system shall implement secure password hashing using bcrypt

2.3 The system shall protect against SQL injection, XSS, and CSRF attacks

2.7 The system shall implement session timeout after 30 minutes of inactivity

3. Reliability

3.1 The system shall maintain a uptime during operational hours

3.2 The system shall provide error handling that preserves user input in case of failures

4. Usability

4.1 The system shall provide a responsive design that works on desktop and mobile devices

4.2 The system shall provide clear error messages and validation feedback

4.3 The system shall provide contextual help and tooltips for complex features

4.4 The system shall maintain consistent navigation and interface patterns

5. Scalability

5.1 The database design shall support increasing data volumes without performance degradation

5.2 The system shall implement caching strategies for frequently accessed data

6. Maintainability

6.1 The system shall implement modular design with clear separation of concerns

6.2 The system shall include comprehensive code documentation

7. Compatibility

7.1 The system shall support major browsers including Chrome, Firefox, Safari, and Edge

7.2 The system shall function properly on devices with varying screen sizes

3.4. System Model

3.4.1. Use Case Model

3.4.1.1. Actor Identification

The proposed system involves the following actors, each with specific roles and interactions:

No	Actor	Description
1	Student	University students who register as service providers on the platform. They create and manage service listings, communicate with clients, deliver services, and receive payments.

2	Client	Individuals or organizations seeking services. They browse the marketplace, place orders, communicate requirements, review deliverables, make payments, and provide ratings and reviews.
3	Adminstrator	System administrators who manage the platform, oversee users and listings, handle disputes, configure system settings, and access platform analytics.
4	Payment system	External payment processing system (Stripe) that handles payment transactions, escrow management, and fund disbursement.
5	Notification system	System component responsible for sending email and in-app notifications to users based on various triggers and events.

Table 3.1 Actors of the proposed system

3.4.1.2. Use Case Diagram and Description

The following use case diagram illustrates the primary interactions between actors and the system:

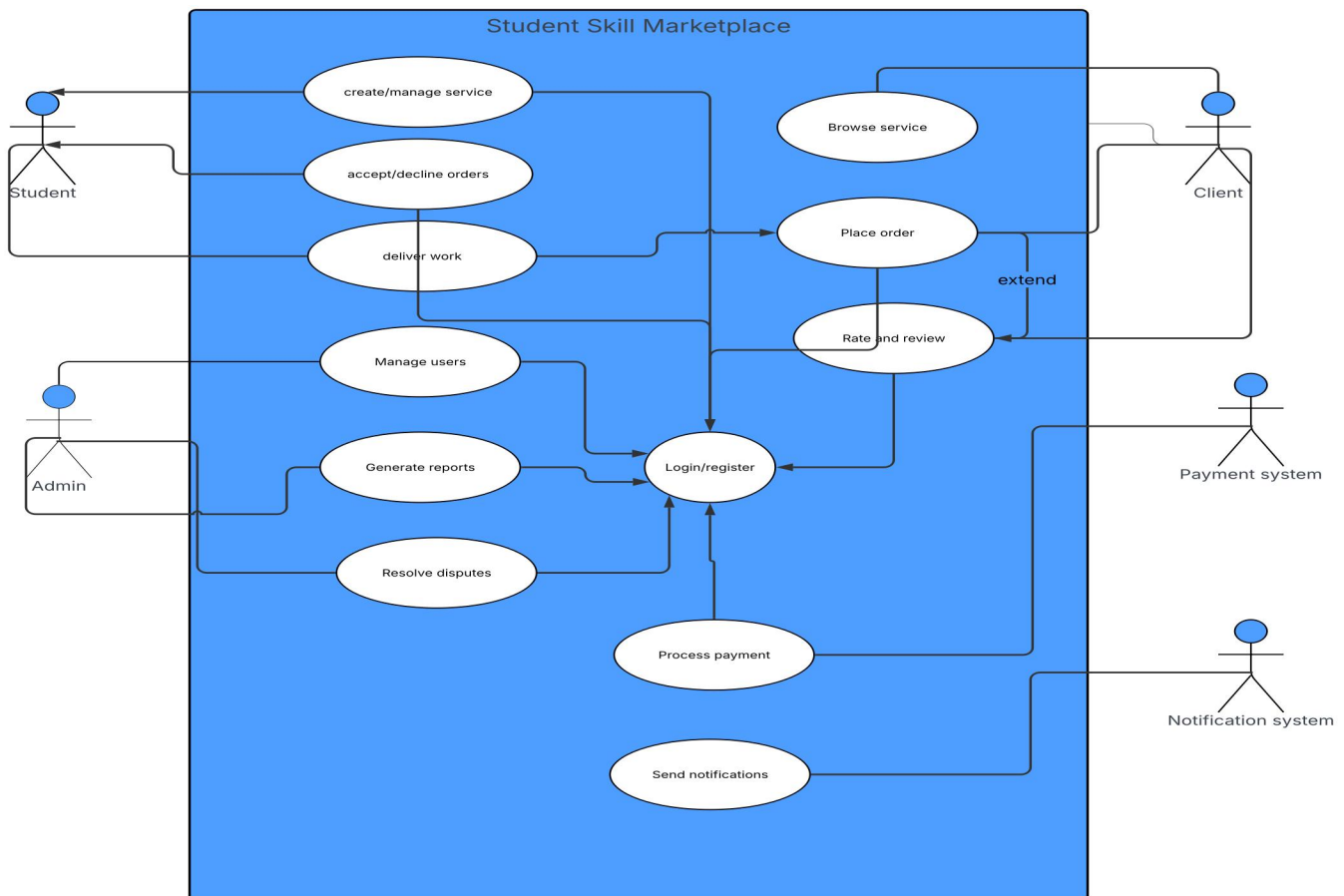


Figure 3.1 Use case diagram

Use Case Description: Register as Student Provider

Attribute	Description
ID	UC-001
Name	Register as Student Provider
Description	A student creates a new account as a service provider on the platform
Actors	Student
Preconditions	Student is not registered in the system
Basic Flow	1. Student navigates to registration page 2. Student selects "Register as Service Provider" 3. Student enters required information (name, email, password) 4. System validates input 5. System creates new account 6. System sends verification email 7. Student verifies email address 8. System activates account
Alternative flows	- Email already registered: System notifies student and suggests login - Validation fails: System displays error messages and allows correction
Postconditions	Student account is created and verified in the system

Table 3.2 Register as service provider usecase description

Use Case Description: Create Service Listing

Attribute	Description
ID	UC-002
Name	Create service listing
Description	A student creates a new service listing to offer to potential clients
Actors	Student
Precondition	Student is logged in and has a verified account
Basic flow	1. Student navigates to "Create New Service" page 2. Student enters service details (title, description, category, tags)

	3. Student sets pricing and delivery timeframes 4. Student uploads sample work (optional) 5. Student submits listing 6. System validates listing information 7. System publishes the service listing
Alternative flows	- Validation fails: System displays error messages and allows correction - Student saves as draft: Listing is saved but not published
Postcondition	New service listing is published and available for clients to discover

Table 3.3 Service creation usecase

Use Case Description: Place Order

Attribute	Description
ID	UC-003
Name	Place order
Description	A client places an order for a student's service
Actors	Client, Payment system
Preconditions	Client is logged in and viewing a service listing
Basic flow	1. Client selects "Order Now" on service listing 2. Client provides project requirements and any necessary files 3. System calculates total price including platform fees 4. Client is redirected to payment processing 5. Client completes payment via Stripe 6. System creates order in "pending" status 7. System notifies student of new order
Alternative flows	- Payment fails: System notifies client and allows retry - Client cancels during process: Order is not created
Postconditions	Order is created, payment is held in escrow, student is notified

Table 3.4 Order placing usecase

Use Case Description: Deliver Completed Work

Attribute	Description
ID	UC-004
Name	Deliver completed work
Description	A student uploads and delivers completed work to fulfill a client's order
Actors	Student
Preconditions	Student has an active order in "in_progress" status
Basic flow	1. Student navigates to order management dashboard 2. Student selects the order to deliver 3. Student uploads completed work files 4. Student adds delivery message with work description 5. Student marks order as "delivered" 6. System notifies client of delivery 7. System updates order status to "delivered" 8. Client receives notification to review the work
Alternative flows	none
post conditions	Order is marked as delivered, client is notified, and payment release process begins

Table 3.5 Deliver completed work use case

Use Case Description: Leave Review and Rating

Attribute	Description
ID	UC-005
Name	Leave review and rating
Description	A client provides feedback and rating for a completed service order
Actors	Client
Preconditions	Client has a completed order and has not yet reviewed the service
Basic flow	1. Client receives notification that order is ready for review 2. Client navigates to completed orders section 3. Client clicks "Leave Review" for the specific order 4. Client selects star rating (1-5 stars)

	5. Client writes detailed review text 6. Client submits the review 7. System validates review content 8. System publishes review on student's profile 9. System updates student's average rating 10. Student receives notification of new review
Alternative flows	- Rating only: Client can submit rating without written review - Review editing: Client can edit review within 24 hours of submission
post conditions	Review is published, student's rating is updated,

Table 3.6 leave review and rating usecase

3.4.2. Static Model

3.4.2.1. Class Diagram

The class diagram represents the structure of the system, showing classes, their attributes, operations, and relationships:

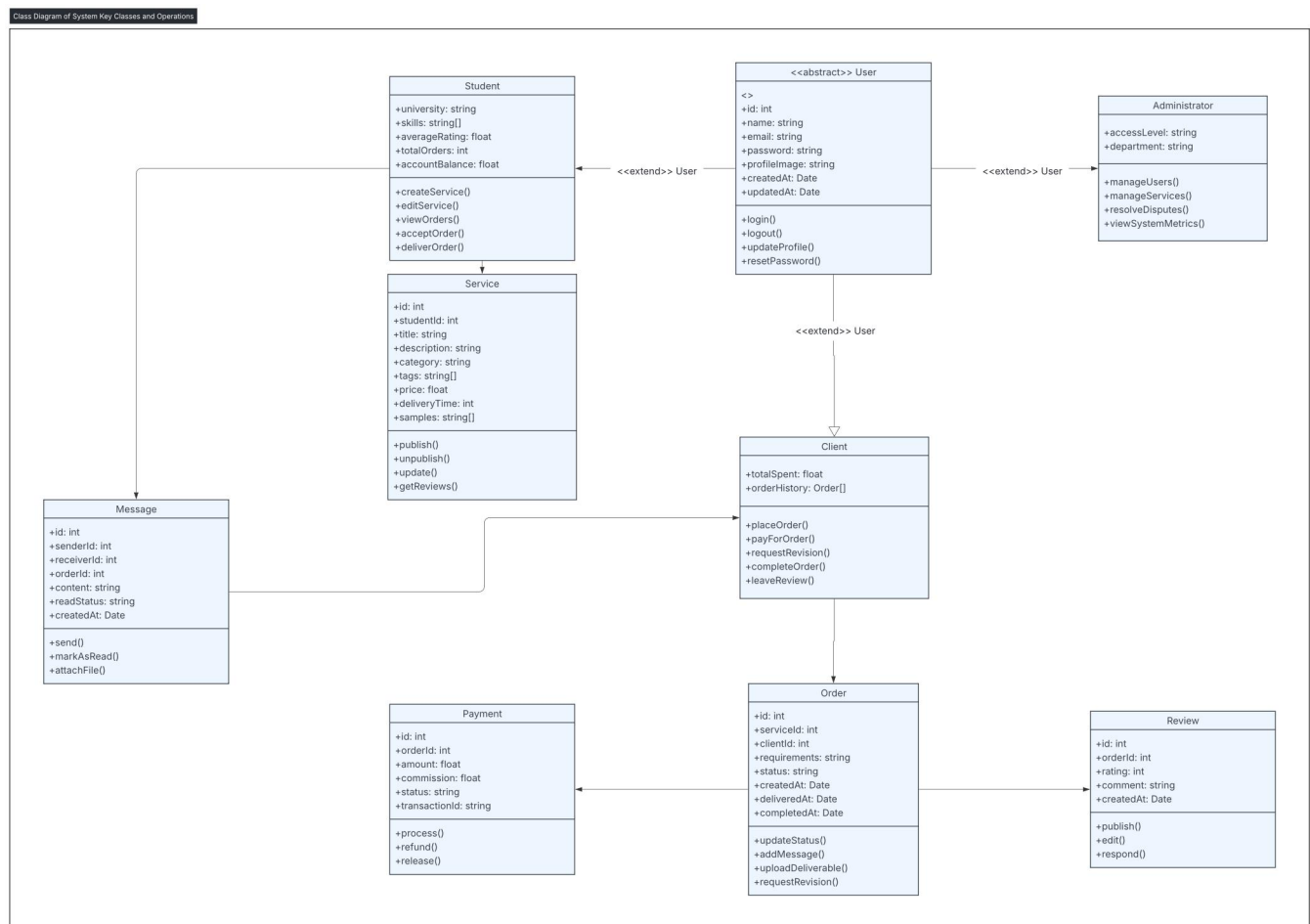


Figure 3.2 Class diagram

3.4.2.2. Object Diagram

The object diagram shows instances of the classes at a specific point in time:

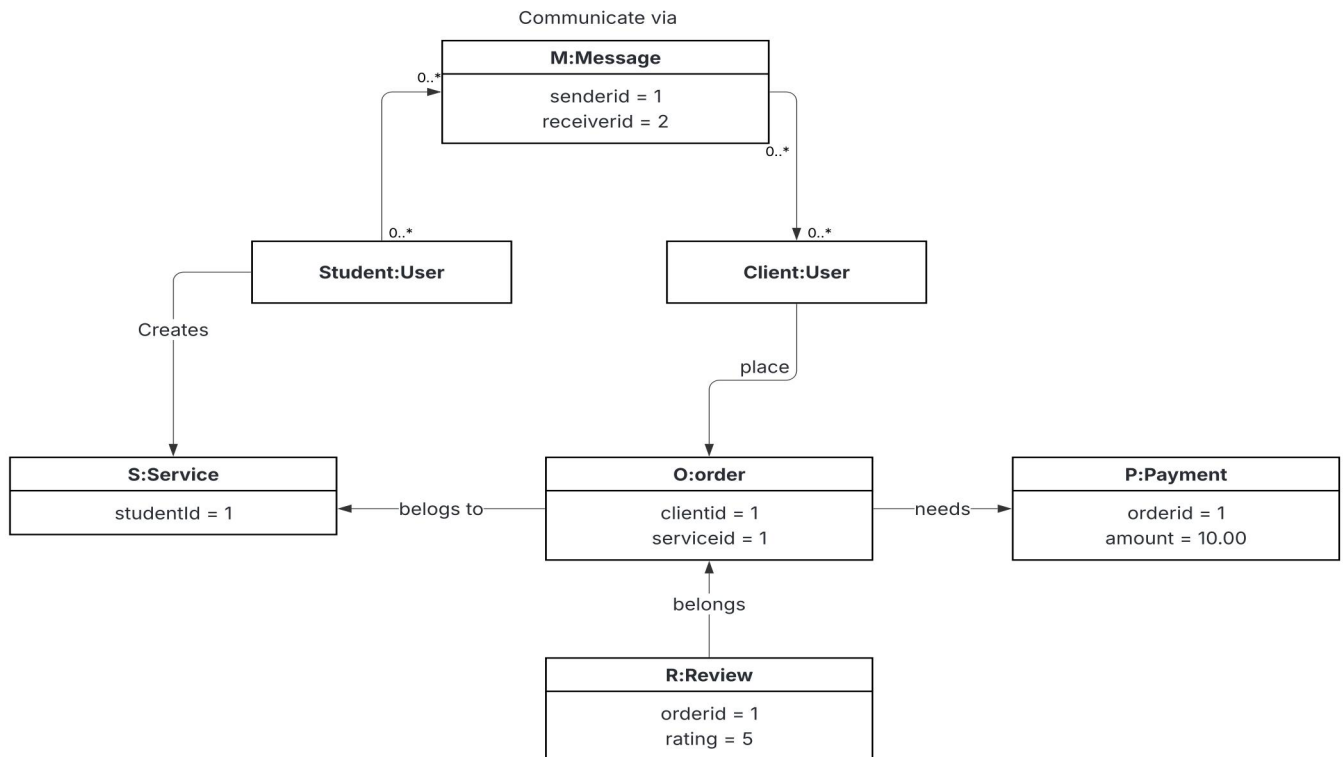


Figure 3.3 Object diagram

3.4.3. Dynamic Model

3.4.3.1. Sequence Diagram

The sequence diagram illustrates the interactions between objects over time for key processes:

User Login Sequence Diagram

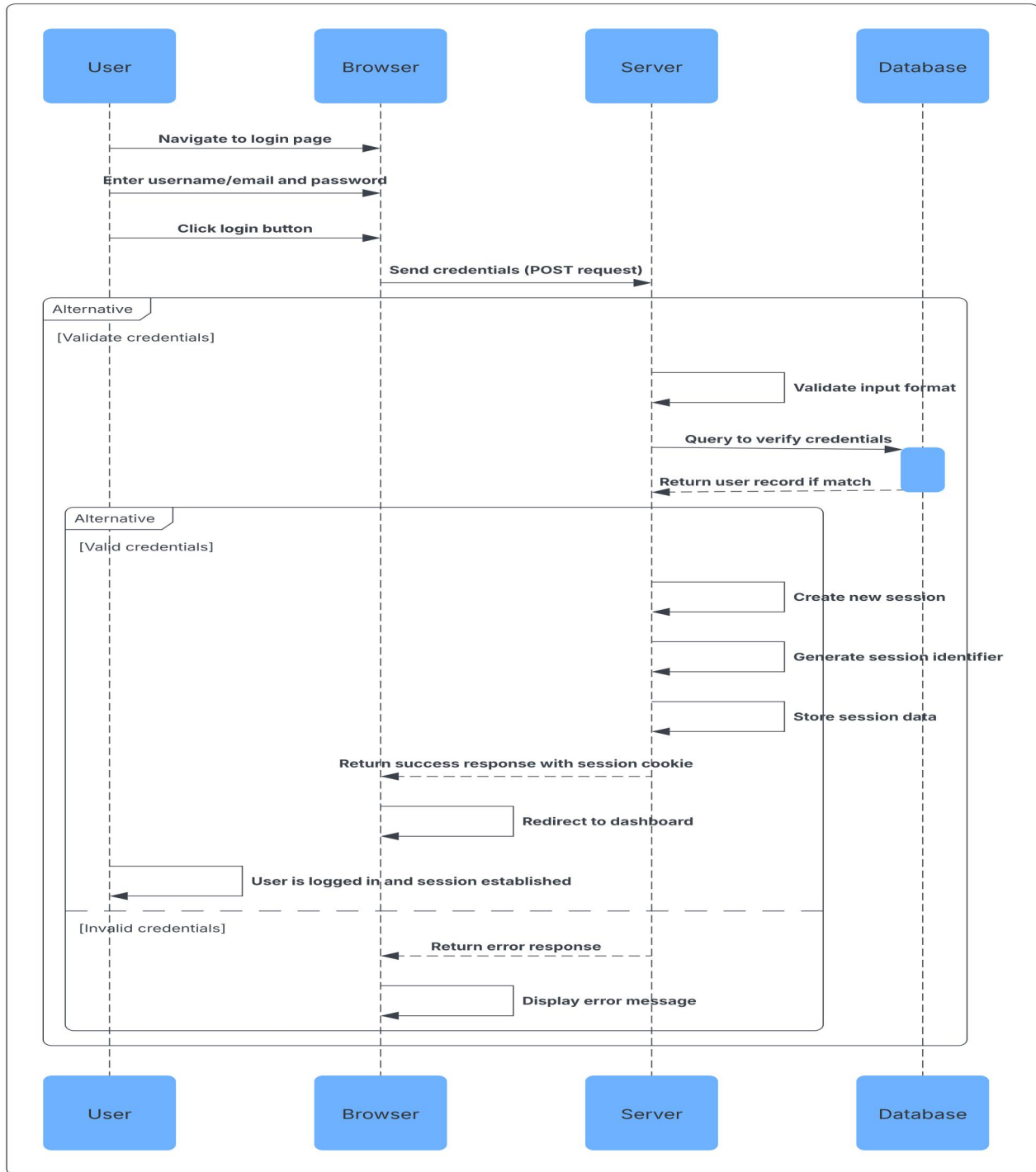


Figure 3.4 Sequence Diagram for Login Use Case

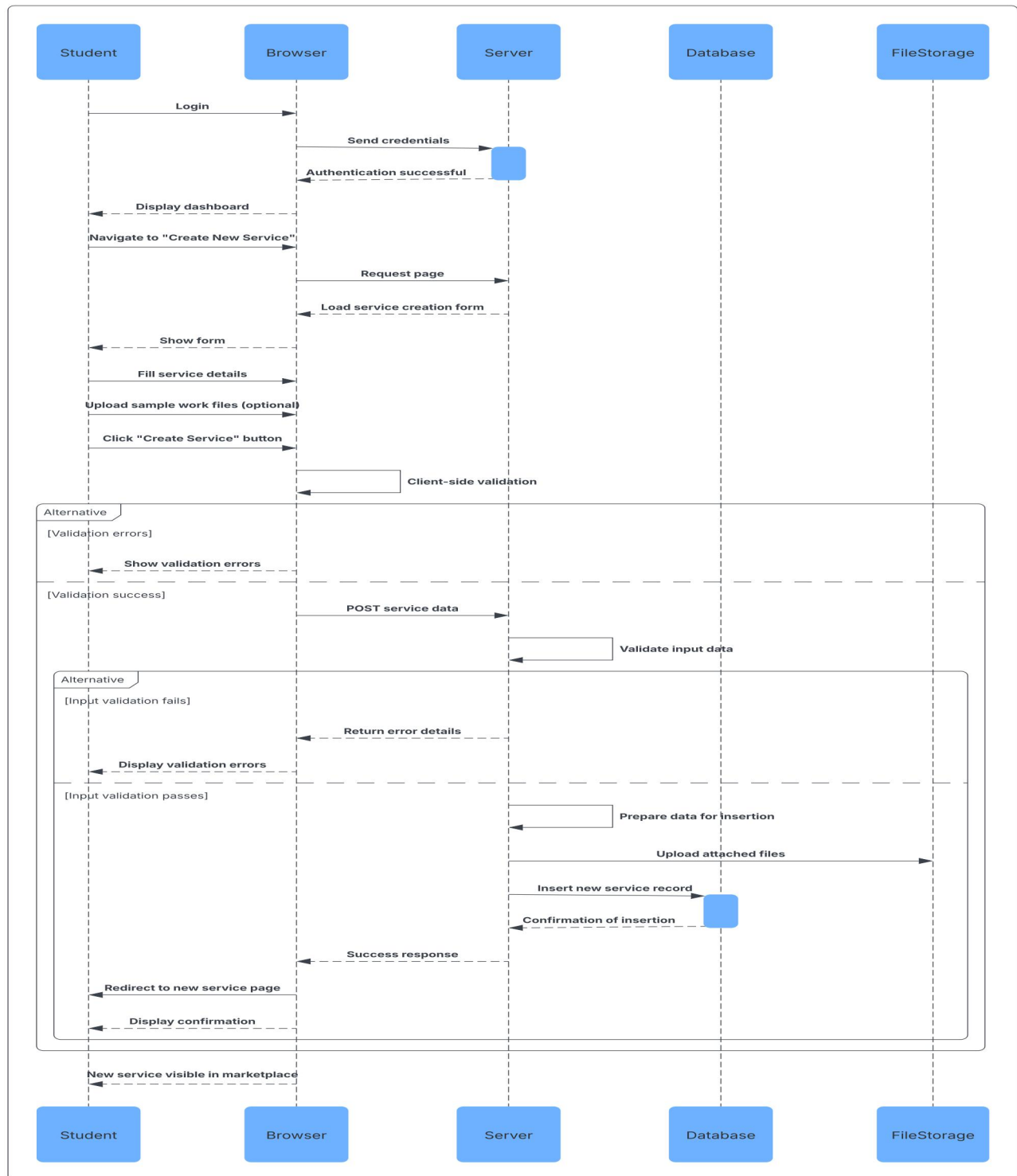


Figure 3.5 sequence diagram for service listing creation

Order Placement and Completion Sequence Diagram

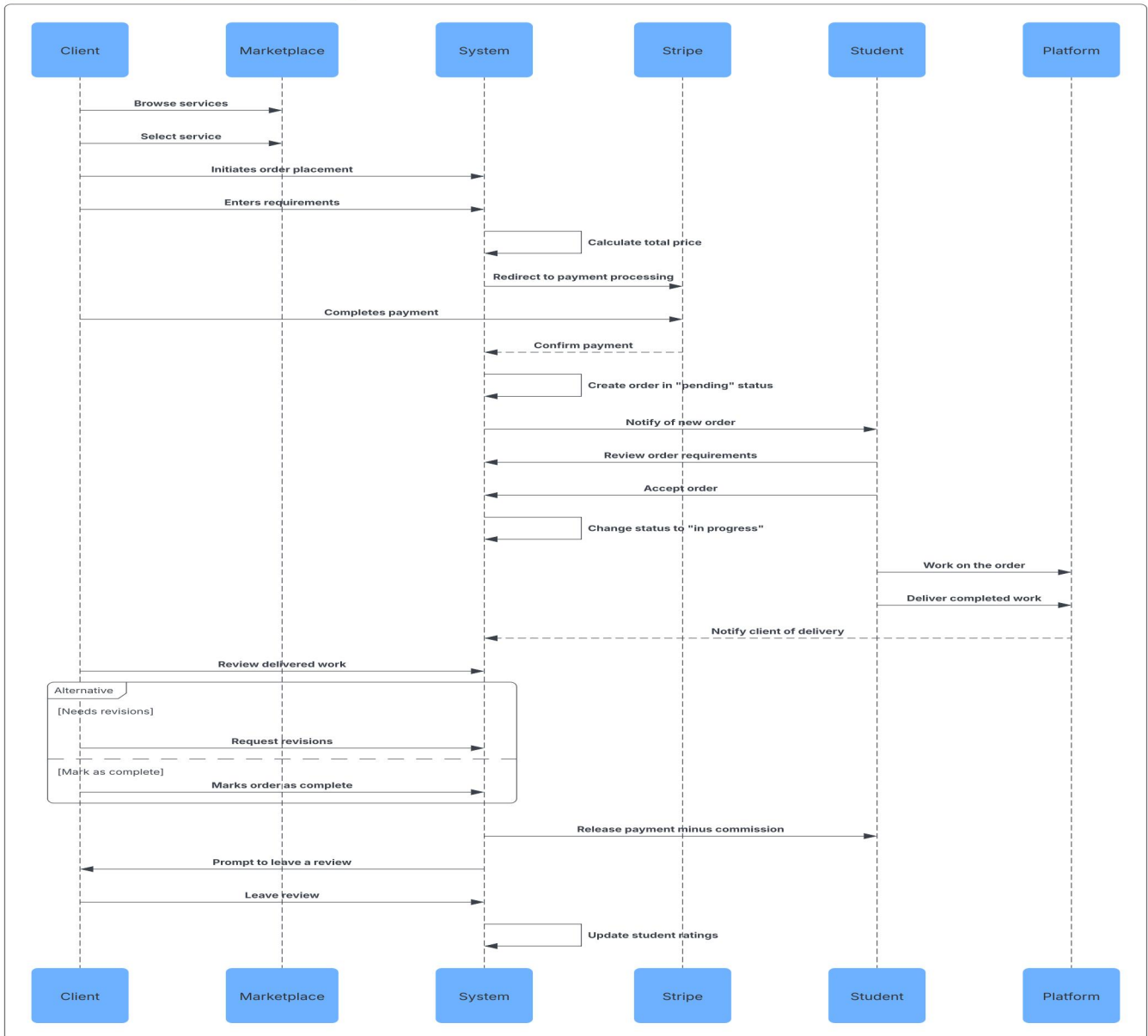


Figure 3.6 sequence diagram for order placement

3.4.3.2. Activity Diagram

This activity diagram represents the complete workflow from service creation through order fulfillment, illustrating the activities performed by students and clients, system processes, and decision points throughout the service lifecycle.

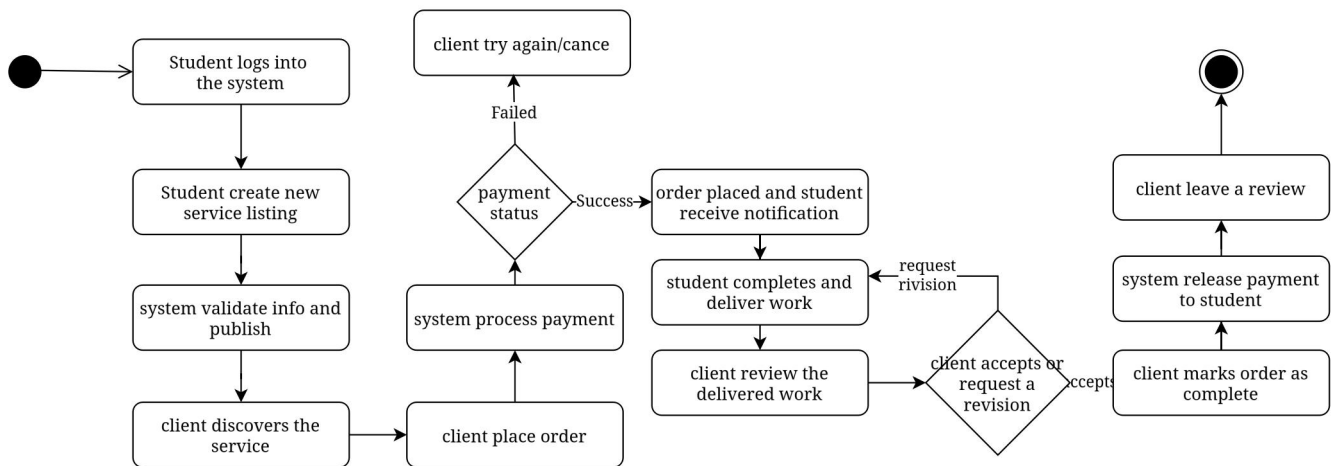


Figure 3.7 activity diagram

3.4.3.3. State Diagram

A UML state diagram is a behavioral modeling diagram that represents the dynamic aspects of a system by illustrating how objects change their state over time in response to events. It belongs to the Unified Modeling Language (UML) specification and serves as a powerful tool for visualizing the lifecycle and behavior of individual objects, components, or entire systems.

State Diagram: Order Lifecycle

This state diagram illustrates the various states an order can exist in throughout its lifecycle within the Student Skill Marketplace system.

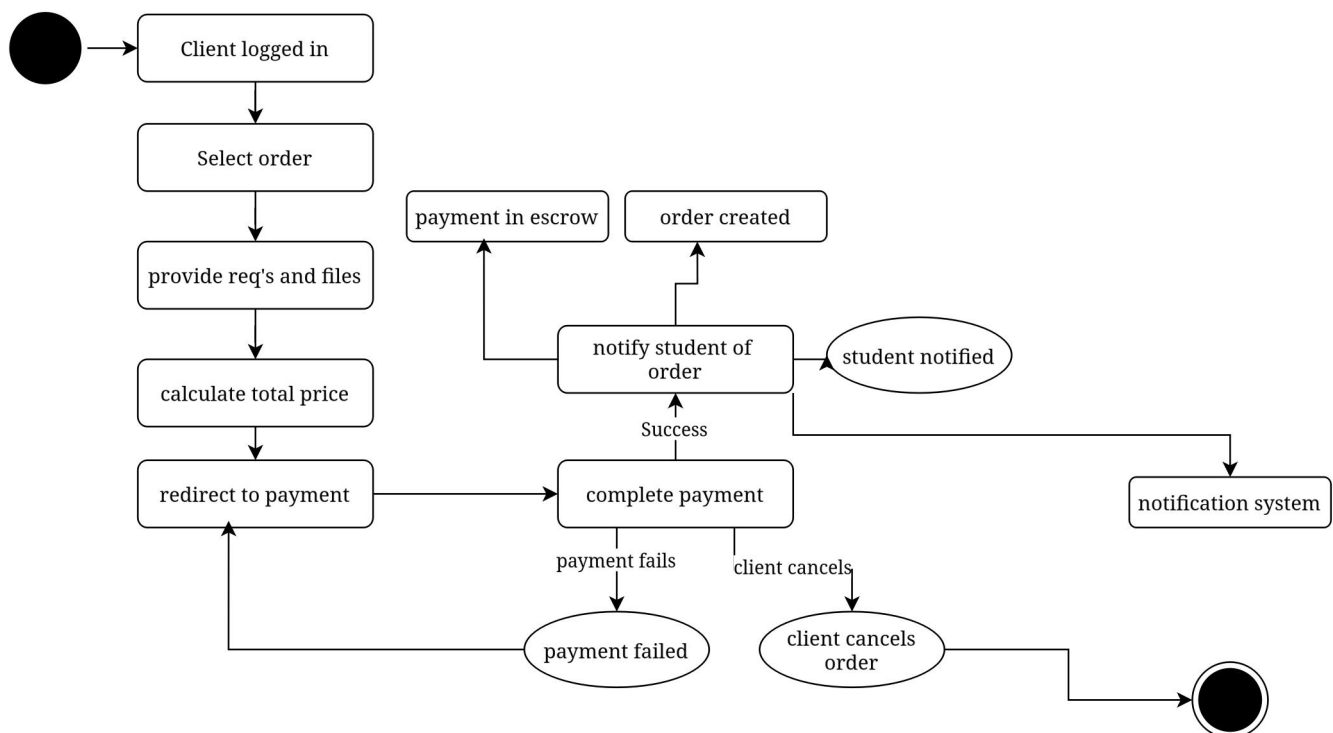


Figure 3.8 state diagram

CHAPTER FOUR: SYSTEM DESIGN

4.1. Overview

The system design phase translates the functional and non-functional requirements identified in the previous chapters into a comprehensive technical blueprint for the Student Skill Marketplace platform. This chapter presents the architectural design, component specifications, data management strategies, and user interface designs that will guide the implementation of the web-based marketplace system.

The design follows a modular, three-tier architecture pattern comprising presentation, application logic, and data persistence layers. This approach ensures maintainability, scalability, and security while supporting the diverse functional requirements of students, clients, and administrators. The system leverages modern web technologies including PHP for server-side processing, MySQL for data persistence, HTML with Tailwind CSS for responsive presentation, and JavaScript for dynamic client-side functionality.

The design incorporates industry-standard security practices, including secure authentication mechanisms, input validation, SQL injection prevention, and encrypted data transmission. Integration with Stripe's payment processing system ensures PCI DSS compliance for financial transactions while maintaining user trust through secure escrow functionality.

4.2. Purpose of the System Design

The primary purpose of this system design is to create a comprehensive technical specification that addresses the identified problems in the current informal student service provision system. The design aims to: Establish a structured platform that standardizes how students offer services and clients engage with providers, replacing the fragmented informal approach with a cohesive digital marketplace. The design specifically addresses the core functional requirements identified in the requirements analysis, including user management, service listing creation and discovery, order processing, secure payment handling, communication facilitation, and administrative oversight.

4.3. Design Goals

The system design is guided by the following strategic goals that ensure the platform meets both immediate functional needs and long-term growth objectives:

Usability and User Experience

Intuitive Navigation: Design user interfaces that require minimal learning curve for students and clients with varying levels of technical expertise.

Responsive Design: Ensure consistent functionality and appearance across desktop, tablet, and mobile devices to accommodate diverse user preferences.

Performance and Reliability

Fast Response Times: Optimize system performance to load pages faster

Efficient Resource Utilization: Implement caching strategies and database optimization to minimize server resource consumption.

Security and Privacy

Data Protection: Encrypt sensitive user information and implement secure data transmission protocols.

Authentication Security: Use industry-standard password hashing and session management practices.

4.4. Proposed System Architecture

The Student Skill Marketplace employs a three-tier client-server architecture that separates presentation, application logic, and data management concerns. This architectural pattern provides flexibility, maintainability, and scalability while ensuring security and performance optimization.

Presentation Tier (Client Layer)

The presentation tier consists of web browsers accessing the system through responsive HTML interfaces styled with Tailwind CSS and enhanced with JavaScript functionality. This tier handles:

User Interface Rendering: HTML templates generate dynamic content based on user roles and system state.

Client-Side Validation: JavaScript validates user input before submission to reduce server load and improve user experience.

Responsive Layout: Tailwind CSS ensures consistent appearance across different device types and screen sizes.

Interactive Elements: JavaScript manages dynamic content updates, form interactions, and real-time notifications.

Application Tier (Business Logic Layer)

The middle tier, implemented in PHP, processes business logic, manages user sessions, and coordinates interactions between the presentation and data layers. Key components include:

Authentication Module: Manages user registration, login, logout, and session security.

Service Management Module: Handles service listing creation, editing, search, and discovery functionality.

Order Processing Module: Manages the complete order lifecycle from placement through completion.

Payment Integration Module: Coordinates with Stripe APIs for secure payment processing and fund management.

Communication Module: Facilitates messaging between students and clients.

Notification Module: Manages email and in-app notifications for system events.

Data Tier (Persistence Layer)

The data tier, implemented using MySQL, manages all persistent data storage and retrieval operations. This layer includes:

User Data Management: Stores student, client, and administrator account information and profiles.

Service Catalog: Maintains service listings, categories, and metadata.

Transaction Records: Persists order, payment, and commission data for financial tracking.

Communication Logs: Stores message history and notification records.

System Configuration: Maintains platform settings and administrative configurations.

External Integration Layer

The system integrates with external services to provide enhanced functionality:

Stripe Payment Gateway: Handles secure payment processing, escrow management, and fund disbursement.

Email Service Provider: Manages email notifications and verification messages.

File Storage Service: Manages uploaded files including service samples and order deliverables.

4.5. System Process

The system operates through interconnected processes that manage the complete lifecycle of marketplace activities. These processes ensure seamless user experiences while maintaining data integrity and security.

User Registration and Authentication Process

Initial Registration: Users select registration type (student or client) and provide required information including email verification.

Email Verification: System sends verification email with secure token to confirm account authenticity.

Profile Setup: Users complete profile information including personal details, skills (for students), and preferences.

Authentication: Secure login process validates credentials and establishes authenticated sessions with appropriate role-based permissions.

Service Listing Process

Service Creation: Students create service listings with title, description, pricing, category, delivery timeframes, and optional sample work.

Content Validation: System validates listing content for completeness, appropriate categorization, and compliance with platform policies.

Publication: Approved listings become searchable and discoverable in the marketplace.

Ongoing Management: Students can edit, deactivate, or delete listings while maintaining order fulfillment obligations.

Order Processing Workflow

Service Discovery: Clients browse, search, and filter services to identify suitable providers.

Order Placement: Clients specify requirements, select service packages, and initiate the ordering process.

Payment Processing: Stripe integration securely processes payment and holds funds in escrow until order completion.

Order Fulfillment: Students receive notifications, review requirements, and deliver completed work through the platform.

Quality Review: Clients review deliverables and may request revisions within defined parameters.

Order Completion: Upon client acceptance, funds are released to students minus platform commission.

Feedback Collection: Both parties can provide ratings and reviews to build marketplace trust.

Communication Process

Message Initiation: Users can initiate conversations related to specific services or orders.

Real-time Messaging: Platform facilitates secure messaging with file attachment capabilities.

Notification Delivery: System sends email and in-app notifications for new messages and important updates.

Message History: All communications are logged and accessible to relevant parties for reference.

4.6. Subsystem Decomposition

The Student Skill Marketplace is decomposed into distinct subsystems, each responsible for specific functional areas while maintaining clear interfaces for inter-subsystem communication.

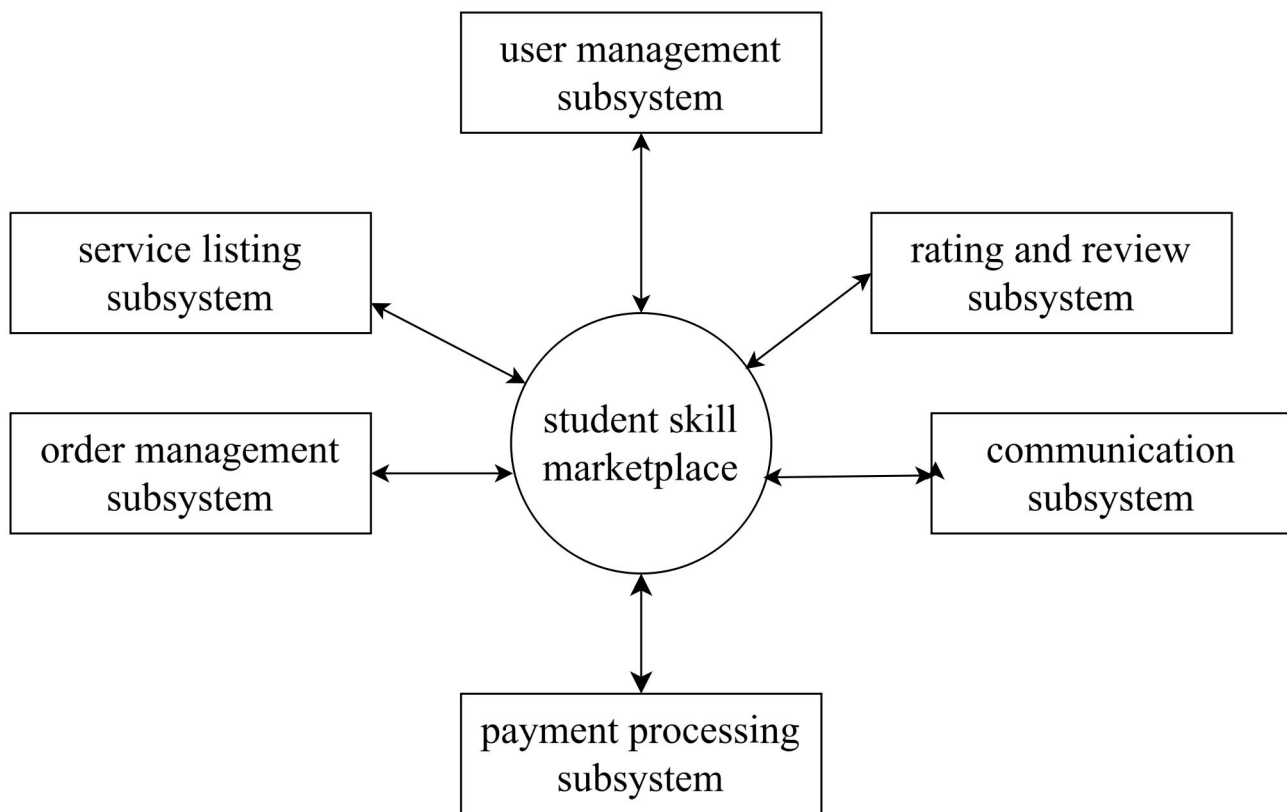


Figure 4.1 system decomposition

4.7. Persistent Data Management

The data management strategy ensures reliable, secure, and efficient storage and retrieval of all system information while maintaining data integrity and supporting analytical requirements.

Database Management System Selection

MySQL has been selected as the primary database management system based on the following considerations:

Reliability: Proven track record in high-transaction web applications

Performance: Optimized for read-heavy workloads typical in marketplace platforms

ACID Compliance: Ensures data consistency for financial transactions

Scalability: Supports horizontal scaling through replication and partitioning

PHP Integration: Native support and extensive documentation for PHP applications

4.8. Component Diagram

The component diagram illustrates the high-level organization of software components and their dependencies within the Student Skill Marketplace system. This diagram provides a clear view of how different parts of the system interact and the flow of data between components.

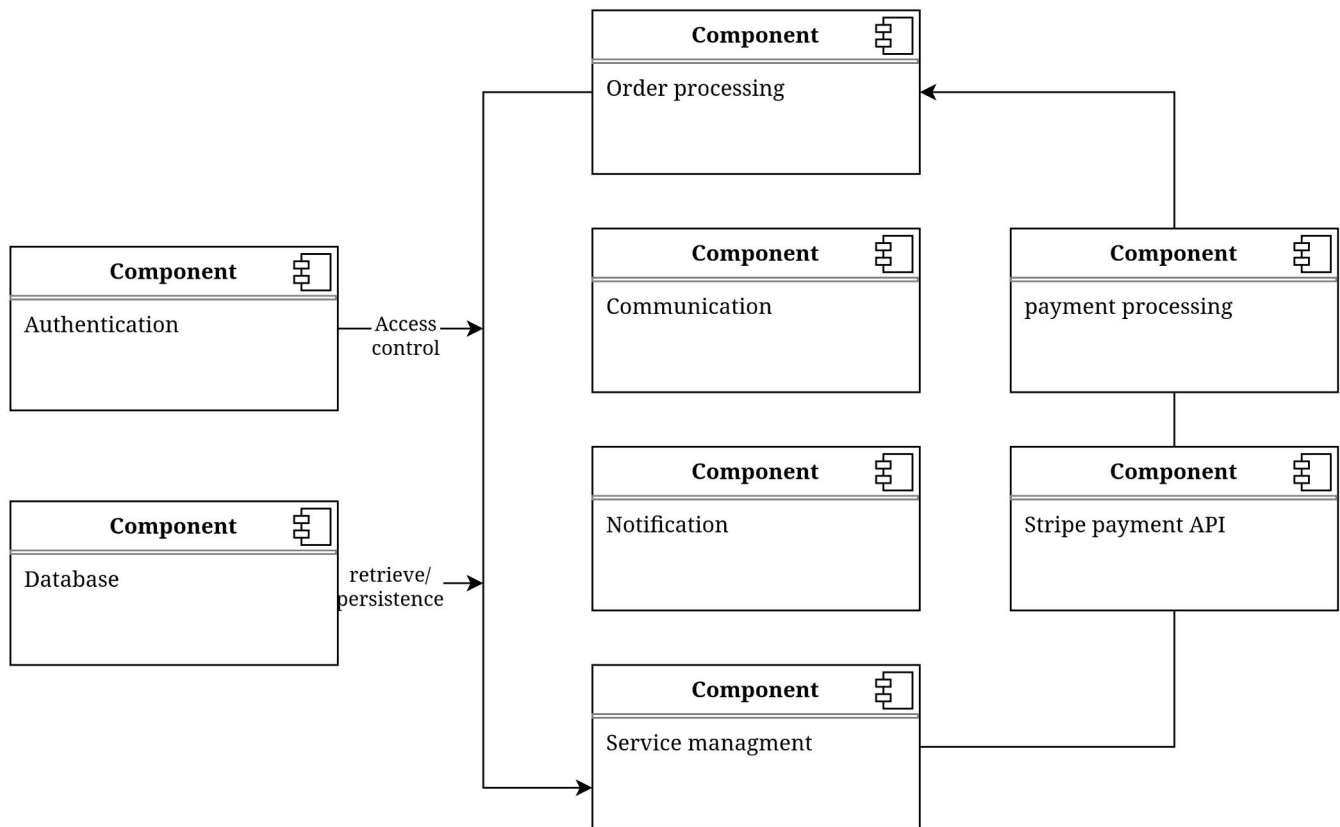


Figure 4.2 component diagram

4.9. Deployment diagram

The deployment diagram shows the physical architecture of the system, including hardware nodes, software components, and network connections required for the Student Skill Marketplace operation.

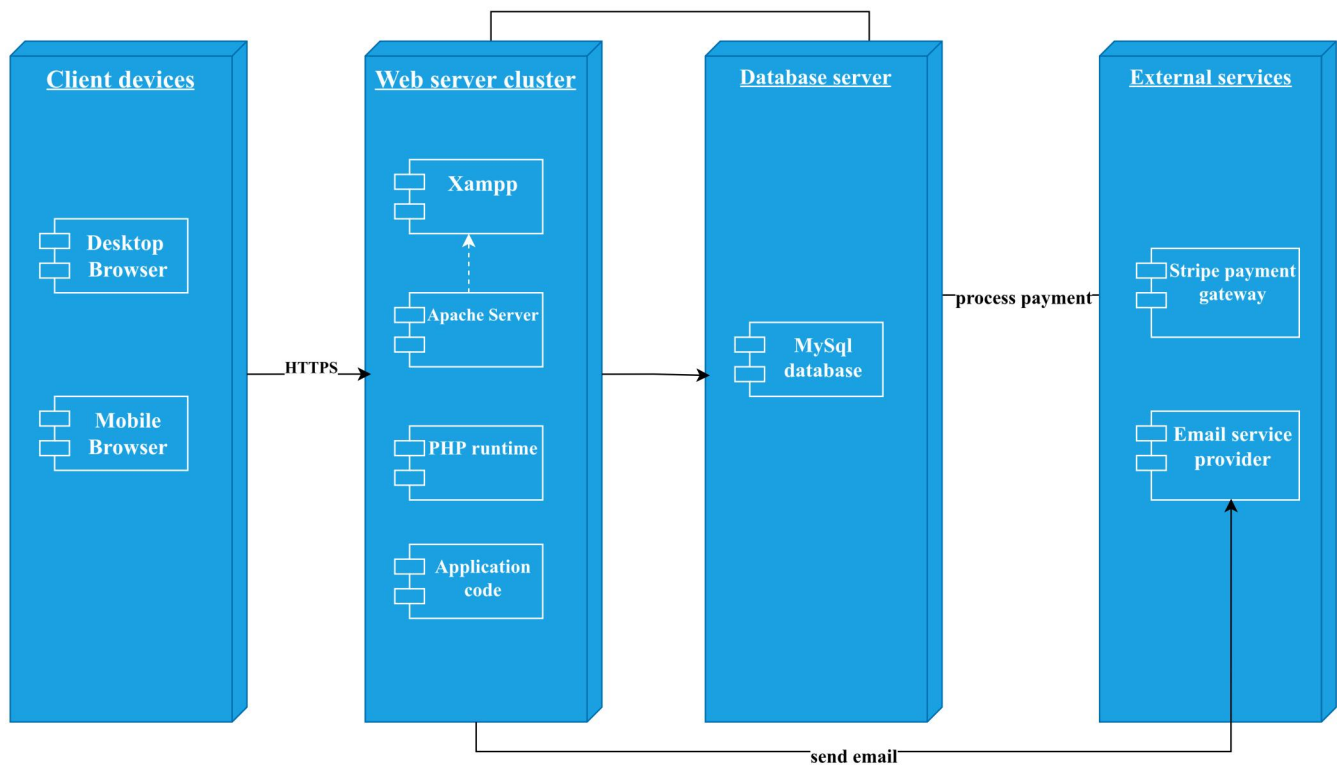


Figure 4.3 deployment diagram

4.10. Database Design

The database design provides the foundation for all data storage and retrieval operations in the Student Skill Marketplace. The design follows relational database principles with careful attention to normalization, performance, and scalability.

4.10.1 Conceptual Database Design

This phase of the database design shows the theoretical design of the data requirement of the database and it is illustrated by Entity Relationship Diagram. Accordingly, the following the following figure shows the ER-Diagram of the proposed system.

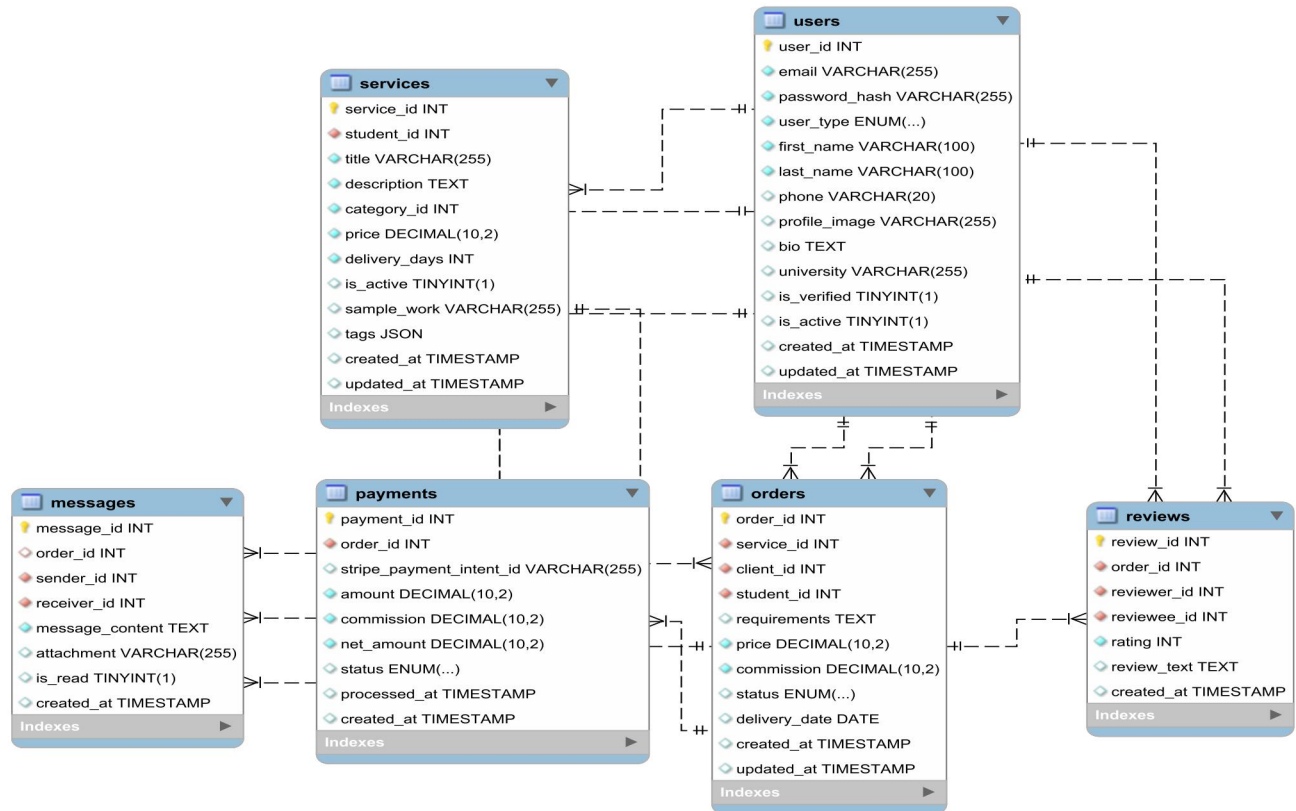


Figure 4.4 Entity Relation Diagram

4.10.2 Physical Database Design

This part of the design shows the physical realization of the data in the database

Column	Type	Default Value	Nullable
bio	text		YES
created_at	timestamp	CURRENT_TIMESTAMP	YES
email	varchar(255)		NO
first_name	varchar(100)		NO
is_active	tinyint(1)	1	YES
is_verified	tinyint(1)	0	YES
last_name	varchar(100)		NO
password_hash	varchar(255)		NO
phone	varchar(20)		YES
profile_image	varchar(255)		YES
university	varchar(255)		YES
updated_at	timestamp	CURRENT_TIMESTAMP	YES
user_id	int		NO
user_type	enum('student','client','admin')		NO

figure 4.5 users table

Column	Type	Default Value	Nullable
category_id	int		NO
created_at	timestamp	CURRENT_TIMESTAMP	YES
delivery_days	int		NO
description	text		NO
is_active	tinyint(1)	1	YES
price	decimal(10,2)		NO
sample_work	varchar(255)		YES
service_id	int		NO
student_id	int		NO
tags	json		YES
title	varchar(255)		NO
updated_at	timestamp	CURRENT_TIMESTAMP	YES

figure 4.6 services table

Column	Type	Default Value	Nullable
client_id	int		NO
commission	decimal(10,2)		NO
created_at	timestamp	CURRENT_TIMESTAMP	YES
delivery_date	date		YES
order_id	int		NO
price	decimal(10,2)		NO
requirements	text		YES
service_id	int		NO
status	enum('pending','in_progress','done')	pending	YES
student_id	int		NO
updated_at	timestamp	CURRENT_TIMESTAMP	YES

figure 4.7 orders table

Column	Type	Default Value	Nullable
attachment	varchar(255)		YES
created_at	timestamp	CURRENT_TIMESTAMP	YES
is_read	tinyint(1)	0	YES
message_content	text		NO
message_id	int		NO
order_id	int		YES
receiver_id	int		NO
sender_id	int		NO

figure 4.8 messages table

Column	Type	Default Value	Nullable
amount	decimal(10,2)		NO
commission	decimal(10,2)		NO
created_at	timestamp	CURRENT_TIMESTAMP	YES
net_amount	decimal(10,2)		NO
order_id	int		NO
payment_id	int		NO
processed_at	timestamp		YES
status	enum('pending','completed','failed','refunded')	pending	YES
stripe_payment_intent_id	varchar(255)		YES

figure 4.9 payments table

Column	Type	Default Value	Nullable
created_at	timestamp	CURRENT_TIMESTAMP	YES
order_id	int		NO
rating	int		NO
review_id	int		NO
review_text	text		YES
reviewee_id	int		NO
reviewer_id	int		NO

figure 4.10 reviews table

4.11. Access Control

The Student Skill Marketplace implements a role-based access control (RBAC) system that restricts user access to features and data based on their assigned roles within the platform. This mechanism ensures security and maintains appropriate boundaries between different user types while providing seamless access to authorized functionality.

Role-Based Access Control Implementation

The system defines three primary user roles, each with specific permissions and access levels:

Student Role Access:

Students can create and manage their service listings, and fulfill orders, and communicate with clients. For example, when a student logs in, they can access the "Create New Service" page, view their active orders in the dashboard, and respond to client messages. However, they cannot access admin functions like user management or view other students' financial data.

Client Role Access:

Clients can browse services, place orders, and manage their purchases. For instance, a client can search for graphic design services, place an order for a logo design, track the order

progress, and leave reviews upon completion. Clients cannot create service listings or access administrative functions.

Administrator Role Access:

Administrators have comprehensive system access including user management, content moderation, and platform analytics. For example, an admin can suspend user accounts, moderate service listings for policy compliance, view platform-wide transaction reports, and configure system settings like commission rates.

4.12. User Interface Design

The user interface design prioritizes usability, accessibility, and responsive design to ensure an optimal experience across all user types and devices.

4.13. References

1. *Martin Fowler, (2002). Patterns of Enterprise Application Architecture.*
2. *Craig Larman, (2004). Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development.*
3. *Sommerville, I. (2015). Software Engineering. 10th Edition. Pearson Education Limited.*